

SMART CPU SCHEDULER

An Interactive Simulation & AI Recommendation Platform

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Subject: Operating Systems

| PROBLEM STATEMENT

EFFICIENCY CHALLENGES

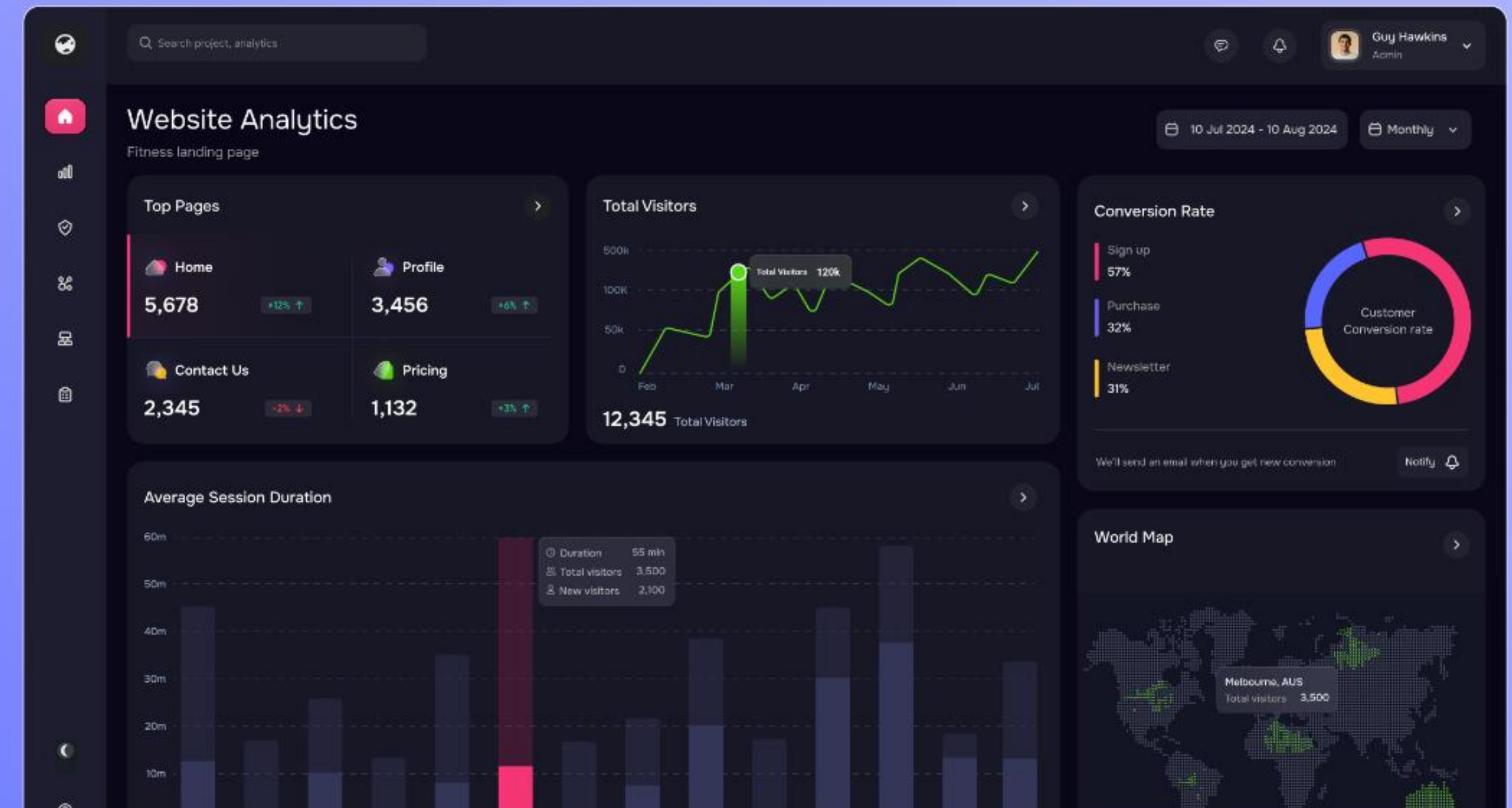
Modern OS execute multiple processes simultaneously. Choosing the **wrong algorithm** leads to high waiting times and low CPU utilization.

THE SOLUTION

A smart simulator that compares algorithms in real-time, providing **data-driven AI recommendations** to optimize performance.

PROJECT OBJECTIVES

-  **Simulate** various CPU Scheduling Algorithms.
-  **Compare** performance across key metrics.
-  **Calculate** Waiting and Turnaround times.
-  **Recommend** the best algorithm using AI logic.
-  **Visualize** execution with Gantt Charts.



TECHNOLOGIES USED

Technology	Purpose
Python	Core Programming Language for logic and data processing.
Tkinter	Graphical User Interface (GUI) development.
Matplotlib	Visualization of Gantt Charts and Performance Graphs.
JSON	Structured data storage for process configurations.
VS Code	Primary Integrated Development Environment (IDE).

| WHAT IS CPU SCHEDULING?

CPU Scheduling is the crucial process of **selecting** which process in the ready queue should be allocated the CPU next.

"Since only one process can use the CPU at a time, the scheduler manages the execution order to maximize efficiency and responsiveness."



SCHEDULING ALGORITHMS

FCFS

First Come First Served: Simple and fair, but can lead to the 'Convoy Effect' (High waiting time).

SJF / SRTF

Shortest Job First: Minimizes average waiting time but requires burst time prediction.

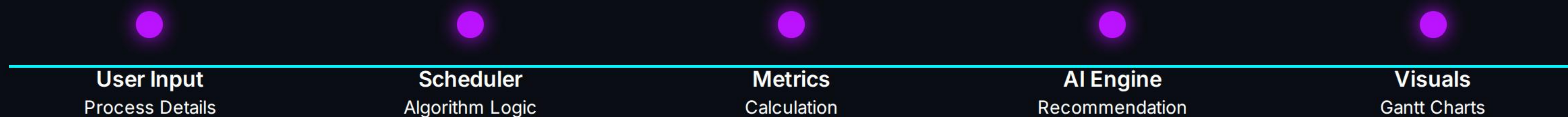
PRIORITY

Priority Based: Executes important tasks first. Risk of 'Starvation' for low priority jobs.

ROUND ROBIN

Time Quantum: Fairly cycles through all processes. Excellent for time-sharing systems.

| SYSTEM WORKFLOW



| INTERACTIVE GUI FEATURES

 Add/Load Process Data

 Real-time Simulation Run

 Reset & Parameter Tuning

 Export Results to PDF

 Save Configuration to JSON

 Dynamic Gantt Chart View

INPUT PARAMETERS & SAMPLE DATA

Each process requires four fundamental values for the simulation:

Process ID	Arrival Time	Burst Time	Priority
P1	0 ms	5 ms	2
P2	1 ms	3 ms	1 (High)
P3	2 ms	6 ms	3 (Low)

| PERFORMANCE METRICS - FORMULAS

WAITING TIME (WT)

$$WT = TAT - BT$$

TURNAROUND TIME (TAT)

$$TAT = CT - AT$$

AT: Arrival Time | BT: Burst Time | CT: Completion Time

SYSTEM EFFICIENCY METRICS

CPU UTILIZATION

$$\text{Util} = \frac{\text{Busy Time}}{\text{Total Time}} \times 100$$

THROUGHPUT

$$\text{Throughput} = \frac{\text{Total Processes}}{\text{Total Execution Time}}$$

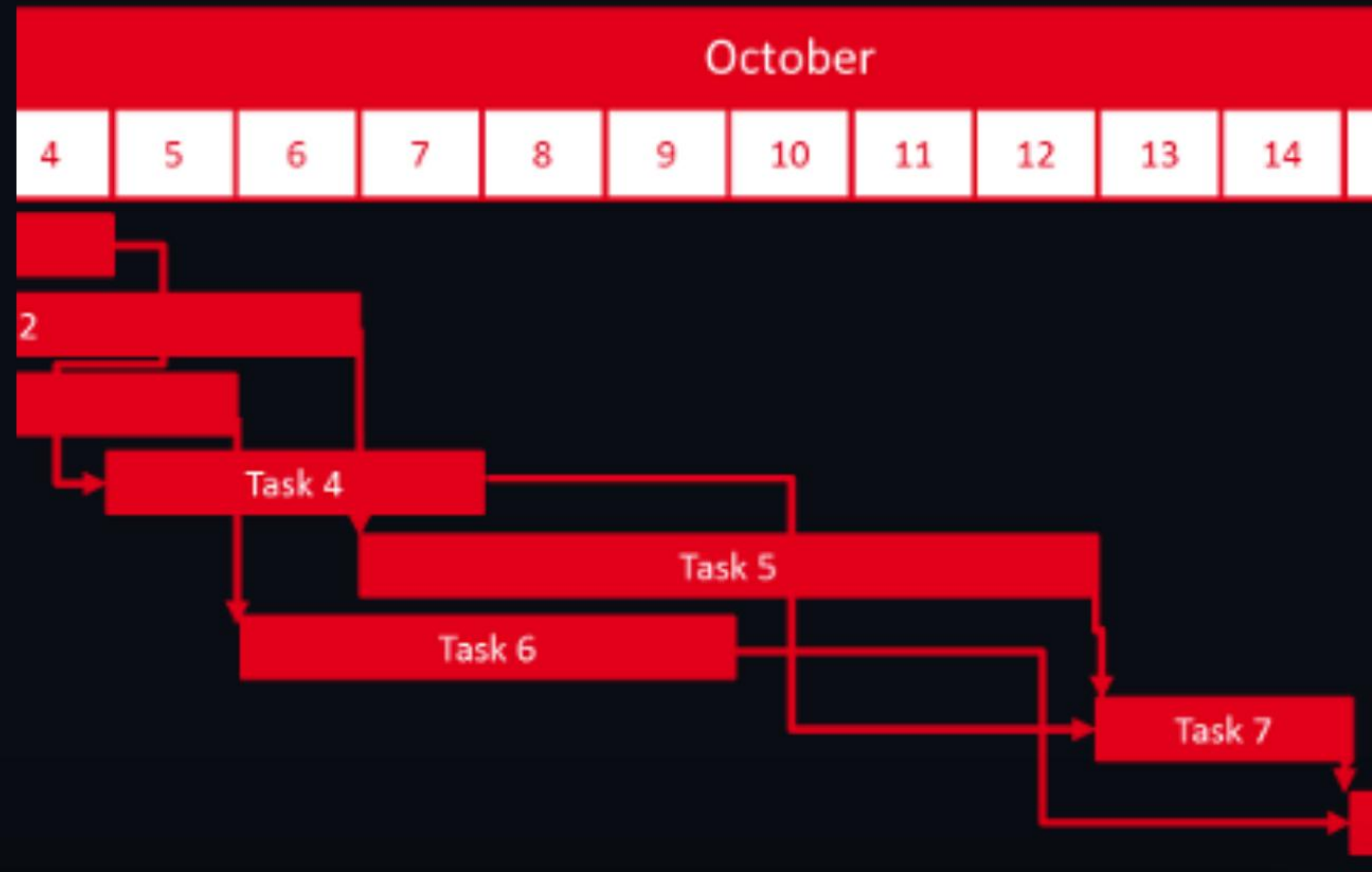
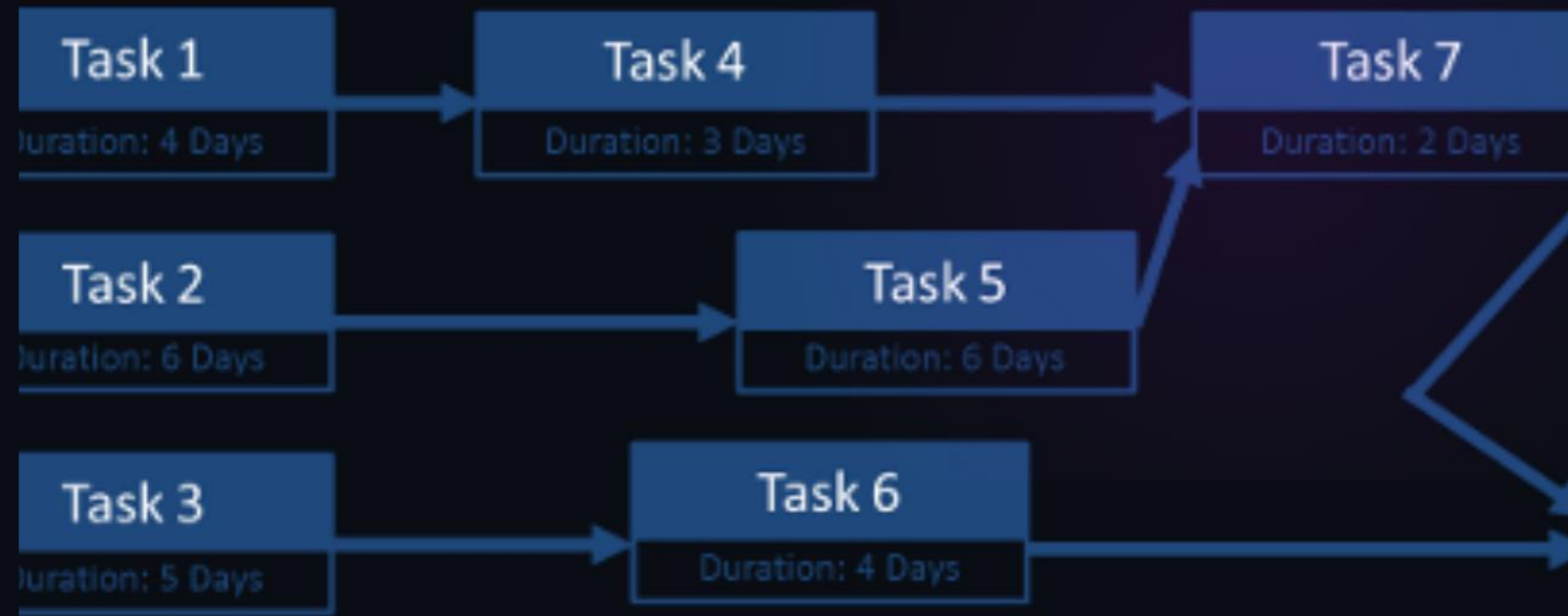
GANTT CHART

The Gantt Chart provides a graphical representation of the process execution sequence over time.

Example Timeline:

| P1 [0-3] | P2 [3-5] | P3 [5-9] |

It allows users to instantly identify execution gaps and overlap efficiency.



AI ALGORITHM RECOMMENDATION



DECISION ENGINE

Analyzes process characteristics like burst time variance, arrival frequency, and priority levels.

High variance? → **SJF**

Time-sharing? → **Round Robin**

Urgent Tasks? → **Priority**

EVALUATION

ADVANTAGES

- User-friendly visualization
- Educational value for students
- Multi-algorithm comparison

LIMITATIONS

- Simulation only (Non-Kernel)
- Predefined rule-based AI
- Limited to simple process types

CONCLUSION & FUTURE SCOPE

The Smart CPU Scheduler offers a robust platform for understanding complex OS concepts through visualization and intelligent recommendation.

FUTURE SCOPE

Machine Learning integration, Cloud-based simulation, and Mobile monitoring app.

- ✓ OS Learning Tool
- ✓ Comparison Engine
- ✓ AI Logic Testing

VIVA PREPARATION / Q&A

Q: Difference between FCFS and RR?

A: FCFS is non-preemptive; RR uses a time quantum for fairness.

Q: What is Starvation?

A: When low-priority tasks never get CPU due to constant high-priority arrivals.

Q: Why SJF is optimal?

A: It mathematically guarantees the minimum average waiting time.

Q: What is the Gantt Chart's role?

A: It visualizes the execution order and duration of processes.

IMAGE SOURCES



<https://cdn.dribbble.com/userupload/43387713/file/original-6af3358e1ae38eab78e7853ab5e95798.png>

Source: dribbble.com



https://img.magnific.com/premium-photo/technology-theme-black-background-with-blue-highlights-futuristic-technological-vibes_163892-4596.jpg?w=360

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<https://blog.netronic.com/hubfs/blog/Gantt%20Chart%20and%20PERT%20Chart%20Comparison%20800.png>

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